

PROJECT 1

Group Project Task List

Student Name	Student ID	Role
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Chapter 1. Introduction

As a group, identify potential problems in your daily life that can be solved using programming.

“Menstrual Cycle Tracker and Reminder”

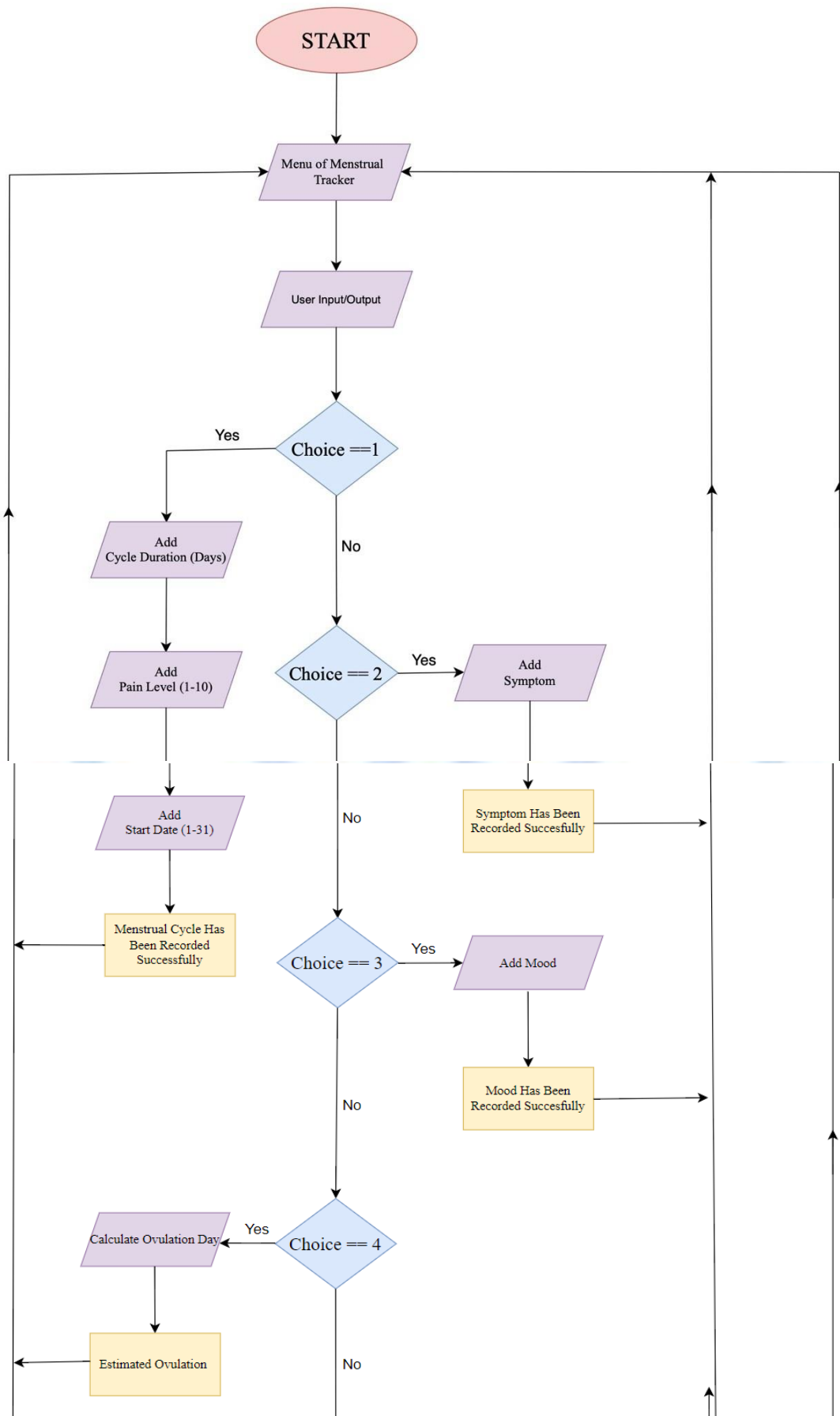
Many women find it difficult to track their menstrual cycles, especially irregular ones. This Menstrual Cycle Tracker and Reminder program aims to help users record how many days their period lasts, predict the next cycle, send reminders before the cycle starts, record symptoms or moods, and read health information related to menstruation.

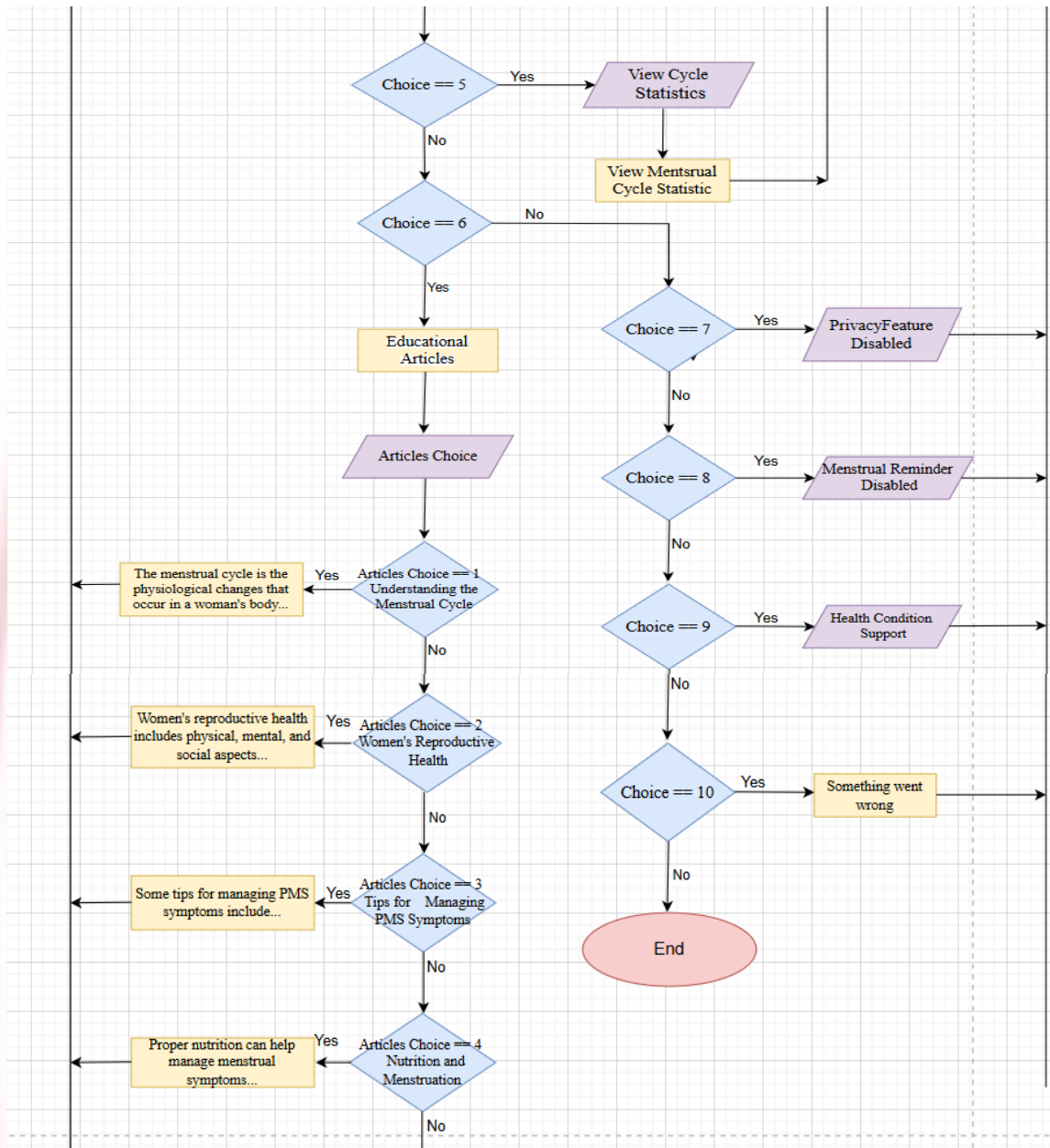
Features:

1. Enable users to record menstrual cycles
2. Enable users to add symptoms
3. Enable users to add moods
4. Enable users to calculate ovulation days
5. Enable users to view cycle statistics
6. Enable users to read health education articles
7. Enable users to enable/disable data privacy
8. Enable users to enable/disable menstrual reminders
9. Enable users to read support for health conditions

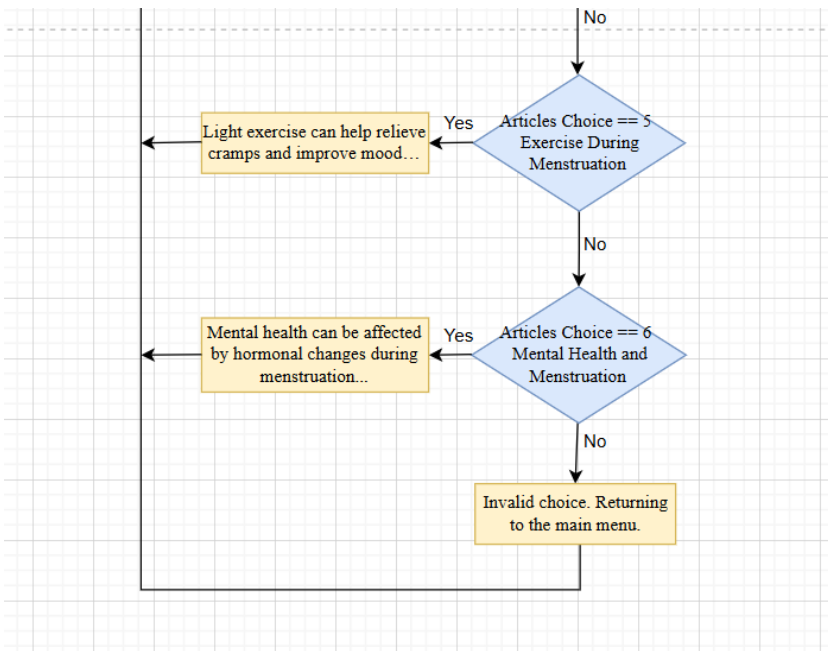
Chapter 2. Flowchart

on the description you provided in Chapter 1, provide the end-to-end flowchart of your program





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Chapter 3: The Program

Provide the program in this section (copy and paste your program here)

Line	Code: “Menstrual Cycle Tracker and Reminder”
1	#include <iostream>
2	#include <string>
3	#include <numeric> // For average calculation
4	#include <iomanip> // For decimal formatting
5	#include <ctime> // For setting reminder dates
6	
7	using namespace std;
8	
9	class MenstrualTracker {
10	private:
11	static const int MAX_RECORDS = 100; // Maximum number of records
12	
13	int cycleDetails[MAX_RECORDS][3]; // Stores cycle details: [duration, pain level,
14	start date]
15	string symptoms[MAX_RECORDS]; // Stores symptoms experienced
16	string moods[MAX_RECORDS]; // Stores mood states
17	
18	int cycleCount; // Tracks the number of recorded cycles
19	int symptomCount; // Tracks the number of recorded symptoms
20	int moodCount; // Tracks the number of recorded moods
21	
22	int lastCycleDuration; // Duration of the last cycle
23	bool dataPrivacyEnabled; // Simulates privacy feature
24	bool reminderEnabled; // Simulates <u>menstrual</u> reminders
25	
26	public:
27	MenstrualTracker()

```

28 : cycleCount(0), symptomCount(0), moodCount(0),
29 lastCycleDuration(28), dataPrivacyEnabled(true), reminderEnabled(true) {}
30
31 // Function to record menstrual cycle
32 void trackCycle(int days, int painLevel, int startDate) {
33     if (cycleCount < MAX_RECORDS) {
34         cycleDetails[cycleCount][0] = days; // Cycle duration
35         cycleDetails[cycleCount][1] = painLevel; // Pain level
36         cycleDetails[cycleCount][2] = startDate; // Start date
37         cycleCount++;
38         lastCycleDuration = days;
39         cout << "Cycle with duration " << days
40             << " days, pain level " << painLevel
41             << ", and start date " << startDate
42             << " has been recorded successfully.\n";
43     } else {
44         cout << "Cycle records are full. Unable to record more.\n";
45     }
46 }
47
48 // Function to add symptoms experienced by the user
49 void addSymptom(const string& symptom) {
50     if (symptomCount < MAX_RECORDS) {
51         symptoms[symptomCount++] = symptom;
52         cout << "Symptom \" " << symptom << "\" has been recorded
53 successfully.\n";
54     } else {
55         cout << "Symptom records are full. Unable to record more.\n";
56     }
57 }
58
59 // Function to add the user's mood
60 void addMood(const string& mood) {
61     if (moodCount < MAX_RECORDS) {
62         moods[moodCount++] = mood;
63         cout << "Mood \" " << mood << "\" has been recorded successfully.\n";
64     } else {
65         cout << "Mood records are full. Unable to record more.\n";
66     }
67 }
68
69 // Function to calculate ovulation day
70 void calculateOvulation() {
71     if (cycleCount == 0) {
72         cout << "No cycle data recorded yet.\n";
73         return;
74     }
75
76     int totalCycleDays = 0;
77     for (int i = 0; i < cycleCount; i++) {
78         totalCycleDays += cycleDetails[i][0];
79     }
80
81     int averageCycle = totalCycleDays / cycleCount;
82     int ovulationDay = averageCycle - 14;
83     cout << "Estimated ovulation day: Day " << ovulationDay << " of the cycle.\n";

```

```

84     }
85
86     // Function to display menstrual cycle statistics
87     void displayStatistics() {
88         if (cycleCount == 0) {
89             cout << "No statistics data to display yet.\n";
90             return;
91         }
92
93         int totalDays = 0;
94         int maxCycle = cycleDetails[0][0];
95         int minCycle = cycleDetails[0][0];
96
97         for (int i = 0; i < cycleCount; i++) {
98             totalDays += cycleDetails[i][0];
99             if (cycleDetails[i][0] > maxCycle) maxCycle = cycleDetails[i][0];
100            if (cycleDetails[i][0] < minCycle) minCycle = cycleDetails[i][0];
101        }
102
103        double averageCycle = static_cast<double>(totalDays) / cycleCount;
104
105        cout << "\n--- Menstrual Cycle Statistics ---\n";
106        cout << "Number of recorded cycles: " << cycleCount << "\n";
107        cout << "Average cycle duration: " << fixed << setprecision(2) << averageCycle
108        << " days.\n";
109        cout << "Longest cycle duration: " << maxCycle << " days.\n";
110        cout << "Shortest cycle duration: " << minCycle << " days.\n";
111    }
112
113
114    // Function to display educational health articles
115    void displayEducationalArticles() {
116        cout << "\n--- Educational Health Articles ---\n";
117        cout << "1. Understanding the Menstrual Cycle\n";
118        cout << "2. Women's Reproductive Health\n";
119        cout << "3. Tips for Managing PMS Symptoms\n";
120        cout << "4. Nutrition and Menstruation\n";
121        cout << "5. Exercise During Menstruation\n";
122        cout << "6. Mental Health and Menstruation\n";
123        cout << "Enter article number to read more (0 to go back): ";
124
125        int articleChoice;
126        cin >> articleChoice;
127
128        switch (articleChoice) {
129            case 1:
130                cout << "\n--- Understanding the Menstrual Cycle ---\n";
131                cout << "The menstrual cycle is the physiological changes that occur in a
132                woman's body...\n";
133                break;
134            case 2:
135                cout << "\n--- Women's Reproductive Health ---\n";
136                cout << "Women's reproductive health includes physical, mental, and
137                social aspects...\n";
138                break;
139            case 3:
140                cout << "\n--- Tips for Managing PMS Symptoms ---\n";

```

141	cout << "Some tips for managing PMS symptoms include...\n";
142	break;
143	case 4:
144	cout << "\n--- Nutrition and <u>Menstruation</u> ---\n";
145	cout << "Proper nutrition can help manage <u>menstrual</u> symptoms...\n";
146	break;
147	case 5:
148	cout << "\n--- Exercise During <u>Menstruation</u> ---\n";
149	cout << "Light exercise can help relieve cramps and improve mood...\n";
150	break;
151	case 6:
152	cout << "\n--- Mental Health and <u>Menstruation</u> ---\n";
153	cout << "Mental health can be affected by <u>hormonal</u> changes during
154	<u>menstruation</u> ...\n";
155	break;
156	case 0:
157	cout << "Returning to the main menu...\n";
158	break;
159	default:
160	cout << "Invalid choice. Returning to the main menu.\n";
161	break;
162	}
163	}
164	
165	// Function to enable or disable the privacy feature
166	void toggleDataPrivacy() {
167	dataPrivacyEnabled = !dataPrivacyEnabled;
168	cout << "Privacy feature " << (dataPrivacyEnabled ? "enabled.\n" :
169	"disabled.\n");
170	}
171	
172	// Function to enable or disable <u>menstrual</u> reminders
173	void toggleReminder() {
174	reminderEnabled = !reminderEnabled;
175	cout << " <u>Menstrual</u> reminder " << (reminderEnabled ? "enabled.\n" :
176	"disabled.\n");
177	}
178	
179	// Function to display health condition support
180	void supportHealthConditions() {
181	cout << "\n--- Health Condition Support ---\n";
182	cout << "1. PCOS (<u>Polycystic</u> Ovary Syndrome)\n";
183	cout << "2. <u>Endometriosis</u> \n";
184	cout << "3. <u>Hormonal</u> Imbalance\n";
185	}
186	
187	// Function to send <u>menstrual</u> reminders if enabled
188	void sendReminder() {
189	if (reminderEnabled) {
190	cout << "\n[Reminder] Your <u>menstrual</u> cycle may start in a few days. Please
191	be prepared.\n";
192	}
193	}
194	
195	
196	// Simple user interface

```

197 void userFriendlyInterface() {
198     cout << "\n--- Menstrual Tracker ---\n";
199     cout << "1. Record menstrual cycle\n";
200     cout << "2. Add symptom\n";
201     cout << "3. Add mood\n";
202     cout << "4. Calculate ovulation day\n";
203     cout << "5. View cycle statistics\n";
204     cout << "6. Read educational health articles\n";
205     cout << "7. Enable/disable data privacy\n";
206     cout << "8. Enable/disable menstrual reminders\n";
207     cout << "9. Support for health conditions\n";
208     cout << "10. Exit\n";
209 }
210 };
211
212 int main() {
213     MenstrualTracker tracker;
214     int choice;
215     int days, painLevel, startDate;
216     string symptom, mood;
217
218     do {
219         tracker.userFriendlyInterface();
220         cout << "Choose an option: ";
221         cin >> choice;
222
223         switch (choice) {
224             case 1:
225                 cout << "Enter cycle duration (days): ";
226                 cin >> days;
227                 cout << "Enter pain level (1-10): ";
228                 cin >> painLevel;
229                 cout << "Enter start date (1-31): ";
230                 cin >> startDate;
231                 tracker.trackCycle(days, painLevel, startDate);
232                 tracker.sendReminder();
233                 break;
234             case 2:
235                 cout << "Enter symptom: ";
236                 cin.ignore();
237                 getline(cin, symptom);
238                 tracker.addSymptom(symptom);
239                 break;
240             case 3:
241                 cout << "Enter mood: ";
242                 cin.ignore();
243                 getline(cin, mood);
244                 tracker.addMood(mood);
245                 break;
246             case 4:
247                 tracker.calculateOvulation();
248                 break;
249             case 5:
250                 tracker.displayStatistics();
251                 break;
252             case 6:

```


254	tracker.displayEducationalArticles();
255	break;
256	case 7:
257	tracker.toggleDataPrivacy();
258	break;
259	case 8:
260	tracker.toggleReminder();
261	break;
262	case 9:
263	tracker.supportHealthConditions();
264	break;
265	case 10:
266	cout << "Exiting the program.\n";
267	break;
268	default:
269	cout << "Invalid choice. Please try again.\n";
270	}
271	} while (choice != 10);
272	
273	return 0;
274	}
275	
276	



Chapter 4: The Screenshot

Give a screenshot of your feature here. Each screenshot should represent 1 feature, as listed in Chapter 1.

1st Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 1
Enter cycle duration (days): 5
Enter pain level (1-10): 7
Enter start date (1-31): 6
Cycle with duration 5 days, pain level 7, and start date 6 has been recorded successfully.

[Reminder] Your menstrual cycle may start in a few days. Please be prepared.
```

2nd Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 2
Enter symptom: Stomach ache and back pain
Symptom "Stomach ache and back pain" has been recorded successfully.
```

3rd Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 3
Enter mood: Feeling sad and angry
Mood "Feeling sad and angry" has been recorded successfully.
```

4th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 4
Estimated ovulation day: Day -9 of the cycle.
```

5th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 5

--- Menstrual Cycle Statistics ---
Number of recorded cycles: 1
Average cycle duration: 5.00 days.
Longest cycle duration: 5 days.
Shortest cycle duration: 5 days.
```

6th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 6

--- Educational Health Articles ---
1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation
Enter article number to read more (0 to go back): 1

--- Understanding the Menstrual Cycle ---
The menstrual cycle is the physiological changes that occur in a woman's body...
```

7th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 6

--- Educational Health Articles ---
1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation
Enter article number to read more (0 to go back): 2

--- Women's Reproductive Health ---
Women's reproductive health includes physical, mental, and social aspects...
```


8th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 6
```

```
--- Educational Health Articles ---
1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation
Enter article number to read more (0 to go back): 3
```

```
--- Tips for Managing PMS Symptoms ---
Some tips for managing PMS symptoms include...
```

9th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 6
```

```
--- Educational Health Articles ---
1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation
Enter article number to read more (0 to go back): 4
```

```
--- Nutrition and Menstruation ---
Proper nutrition can help manage menstrual symptoms...
```

10th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 5

--- Exercise During Menstruation ---

Light exercise can help relieve cramps and improve mood...

11th Feature Code

--- Menstrual Tracker ---

1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit

Choose an option: 6

--- Educational Health Articles ---

1. Understanding the Menstrual Cycle
2. Women's Reproductive Health
3. Tips for Managing PMS Symptoms
4. Nutrition and Menstruation
5. Exercise During Menstruation
6. Mental Health and Menstruation

Enter article number to read more (0 to go back): 6

--- Mental Health and Menstruation ---

Mental health can be affected by hormonal changes during menstruation...

12th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 7
Privacy feature disabled.
```

13th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 8
Menstrual reminder disabled.
```

14th Feature Code

```
--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 9

--- Health Condition Support ---
1. PCOS (Polycystic Ovary Syndrome)
2. Endometriosis
3. Hormonal Imbalance
```



```

--- Menstrual Tracker ---
1. Record menstrual cycle
2. Add symptom
3. Add mood
4. Calculate ovulation day
5. View cycle statistics
6. Read educational health articles
7. Enable/disable data privacy
8. Enable/disable menstrual reminders
9. Support for health conditions
10. Exit
Choose an option: 10
Exiting the program.

```

Chapter 5: Requirement checks

No	Requirements	Yes/No, Line No X - Y
1	Have at least one selection.	Yes If-else Line 33 Switch Line 128-159
2	Have at least one loop	Yes For loop Line 77-79, 97-101 Do-while Line 218-271
3	Have at least one 1 Dimensional Array	Yes Line 15-16, 51, 62
4	Have at least one 2 Dimensional Array	Yes Line 13, 34-36
5	Have at least one mathematical operator calculation	Yes Addition Line 78 Division Line 81 Subtraction Line 82
6	Have at least one Boolean expression	Yes Checking array bounds Line 33 Toggling features Line 167, 174 Condition Line 271

Chapter 6: Log Hours

<i>Student Name</i>	<i>Activity</i>	<i>Log Hours</i>
Tentin Nofa Tuzika	- Discussion/Planning - Report/Documentation - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 30 minutes - 30 minutes - 30 minutes - 50 minutes
Tessalonika Angeline Purba	- Discussion/Planning - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 30 minutes - 20 minutes - 1 hour
Qwyn Celine Djimondo	- Discussion/Planning - Flowchart - Code first and second feature (1-Dimensional Array statement, Switch statement, User input, Mathematical operator calculation, Print output)	- 30 minutes - 20 minutes - 50 minutes